

Verification

o2_H_Simulator_FVM_1D_Incompressible_Heterogeneous

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1 Objective

The purpose of this verification exercise is to demonstrate that the numerical simulator correctly reproduces the analytical solution for one-dimensional incompressible single-phase flow in a heterogeneous porous medium.

2 Governing Equation

For steady-state incompressible flow with no gravity and no source terms, conservation of mass combined with Darcy's law gives

$$\frac{\partial}{\partial x} \left(-\frac{k(x)}{\mu} \frac{\partial P}{\partial x} \right) = 0, \quad (1)$$

where

- P is pressure [Pa]
- k is permeability [m^2]
- μ is fluid viscosity [Pa·s]

The permeability field consists of two homogeneous regions,

$$k(x) = \begin{cases} k_1, & 0 \leq x < L_1, \\ k_2, & L_1 \leq x \leq L_x, \end{cases} \quad (2)$$

where L_1 is the location of the interface between the two permeability zones and L_x is the total domain length. The boundary conditions are

$$P(0) = P_L, \quad P(L_x) = P_R. \quad (3)$$

3 Analytical Solution

Because the flow is steady state and incompressible, the Darcy velocity is constant throughout the domain,

$$u = -\frac{k_1}{\mu} \frac{dP}{dx} = -\frac{k_2}{\mu} \frac{dP}{dx}. \quad (4)$$

The pressure solution is therefore piecewise linear,

$$P(x) = P_L - \frac{\mu u}{k_1} x, \quad 0 \leq x \leq L_1, \quad (5)$$

and

$$P(x) = P(L_1) - \frac{\mu u}{k_2} (x - L_1), \quad L_1 \leq x \leq L_x. \quad (6)$$

4 Effective Permeability

For steady-state incompressible flow, Darcy's law in one dimension is

$$Q = -\frac{kA}{\mu} \frac{dP}{dx}, \quad (7)$$

where Q is the volumetric flow rate. Since the flow is steady state and there are no sources or sinks, Q is constant throughout the domain. Consider a domain composed of two regions in series,

$$L_x = L_1 + L_2, \quad (8)$$

with permeabilities k_1 and k_2 , respectively. The pressure drop across each region is

$$\Delta P_1 = \frac{\mu Q L_1}{A k_1}, \quad (9)$$

and

$$\Delta P_2 = \frac{\mu Q L_2}{A k_2}. \quad (10)$$

The total pressure drop across the domain is therefore

$$P_L - P_R = \Delta P_1 + \Delta P_2 = \frac{\mu Q}{A} \left(\frac{L_1}{k_1} + \frac{L_2}{k_2} \right). \quad (11)$$

An equivalent homogeneous permeability, k_{eff} , can be defined such that

$$Q = \frac{k_{\text{eff}} A}{\mu} \frac{P_L - P_R}{L_x}. \quad (12)$$

Substituting the expression for the total pressure drop yields

$$\frac{L_x}{k_{\text{eff}}} = \frac{L_1}{k_1} + \frac{L_2}{k_2}, \quad (13)$$

and therefore

$$k_{\text{eff}} = \frac{L_x}{\frac{L_1}{k_1} + \frac{L_2}{k_2}}. \quad (14)$$

For the special case in which the two regions have equal length,

$$L_1 = L_2 = \frac{L_x}{2}, \quad (15)$$

the effective permeability reduces to

$$k_{\text{eff}} = \frac{2k_1 k_2}{k_1 + k_2}, \quad (16)$$

which is the harmonic average of the two permeabilities. The numerical solution obtained using the heterogeneous permeability field can therefore be compared against a homogeneous simulation using k_{eff} . Agreement between the two flow rates validates the implementation of heterogeneous permeability and transmissibility calculations.

5 Simulation Parameters

The simulation parameters are given in Table 1.

Parameter	Symbol	Value
Number of cells	N_x	100
Domain length	L_x	1 m
Zone 1 length	L_1	0.5 m
Permeability 1	k_1	1 m ²
Permeability 2	k_2	5 m ²
Viscosity	μ	1 Pa·s
Left pressure	P_L	10×10^6 Pa
Right pressure	P_R	0 Pa

Table 1: Simulation parameters.

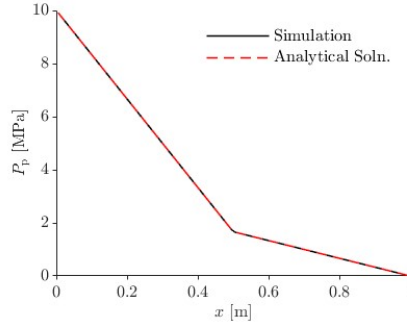


Figure 1: Comparison between analytical and numerical pressure solutions for the MATLAB simulator.

6 Results

6.1 Pressure Distribution

The pressure solutions for both the MATLAB simulator, Figure 1, and the Python simulator, Figure 2, match well.

6.2 Darcy Flux

The Darcy flux solutions for both the MATLAB simulator, Figure 3, and the Python simulator, Figure 4, match well.

7 Error Analysis

The maximum pressure relative error is defined as

$$E_P = \max_i \left| \frac{P_{\text{num},i} - P_{\text{ana},i}}{P_{\text{ana},i}} \right|. \quad (17)$$

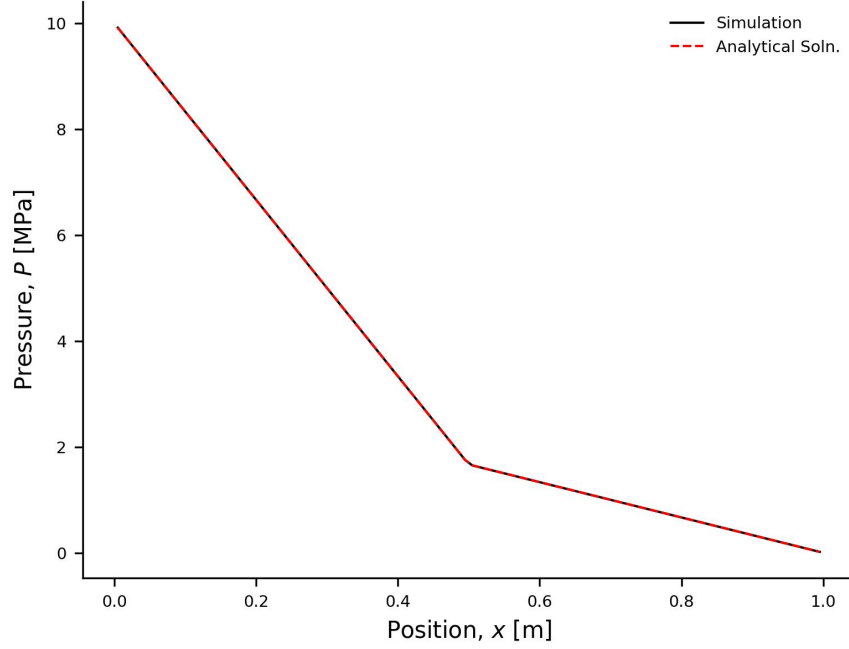


Figure 2: Comparison between analytical and numerical pressure solutions for the Python simulator.

The maximum flux relative error is defined as

$$E_q = \max_i \left| \frac{q_{\text{num},i} - q_{\text{ana},i}}{q_{\text{ana},i}} \right|. \quad (18)$$

For the present simulation in MATLAB

$$E_P = 4.60 \times 10^{-14}, \quad E_q = 5.21 \times 10^{-14}. \quad (19)$$

In Python, the result is

$$E_P = 3.55 \times 10^{-14}, \quad E_q = 3.96 \times 10^{-14}. \quad (20)$$

8 Discussion

The numerical pressure distribution reproduces the analytical linear pressure solution and the numerical Darcy flux reproduces the analytical constant flux solution. The small residual errors arise from floating-point arithmetic and are negligible.

9 Conclusion

The simulator successfully reproduces the analytical solution for one-dimensional incompressible flow in a heterogeneous porous medium. This provides verification that the finite volume implementation is correctly solving the governing equations for this benchmark problem.

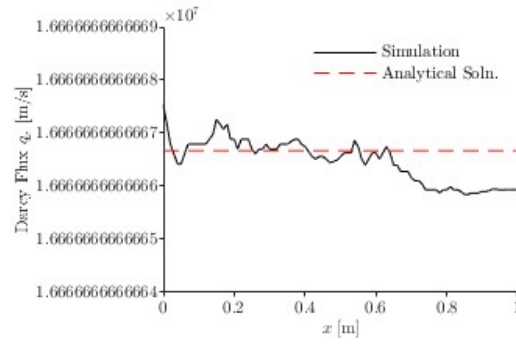


Figure 3: Comparison between analytical and numerical Darcy flux solutions for the MATLAB simulator.

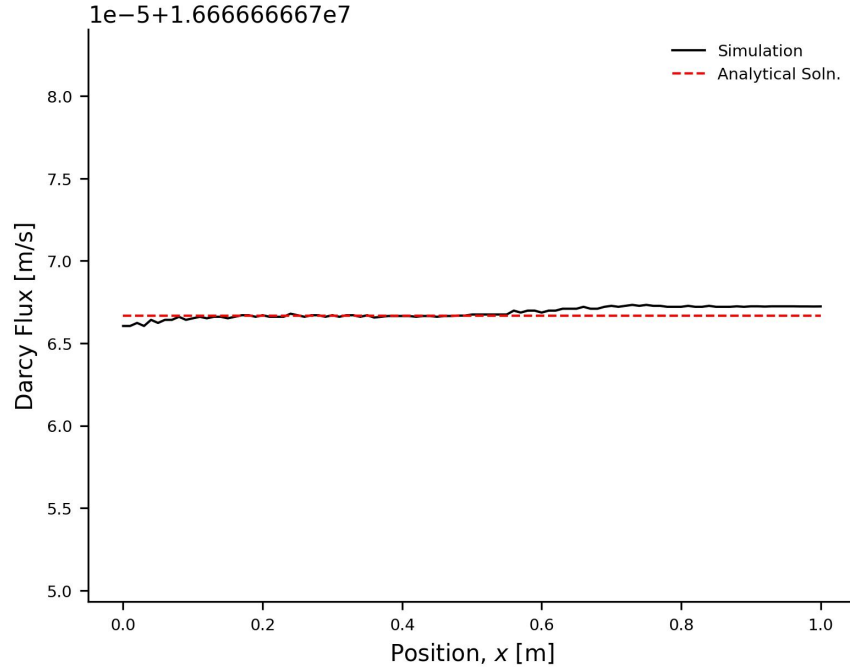


Figure 4: Comparison between analytical and numerical Darcy flux solutions for the Python simulator.